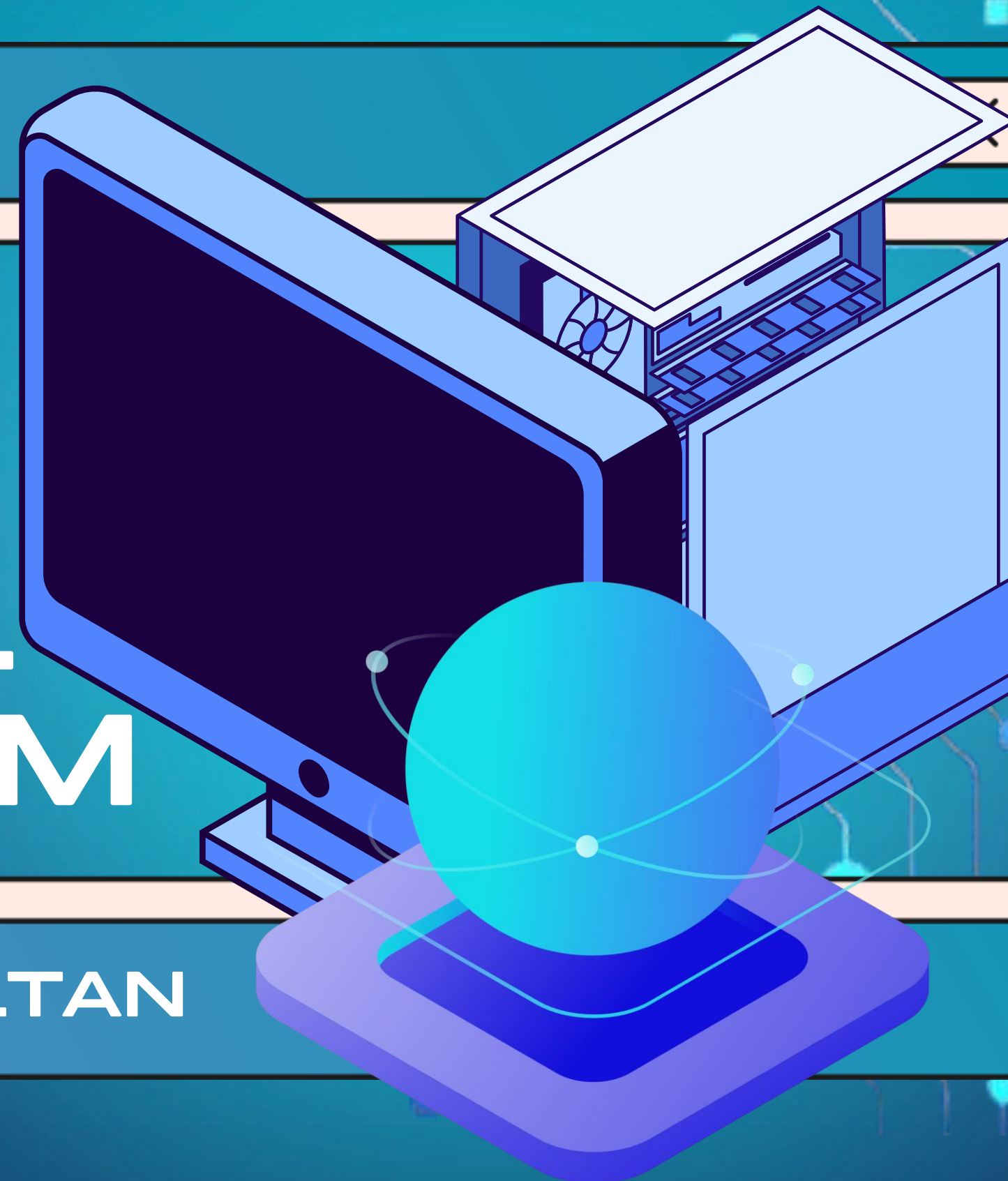


FINAL DEFENSE

AN EXPERT GUIDED MULTIMODAL AI ECOSYSTEM

FARHAN KASHIF, AHMED SULTAN



THE PROBLEM



- **Model Rigidity:** Current models fail when tumor types change
- **Catastrophic Forgetting:** Retraining on new data erases knowledge of old data.
- **Workflow Fragmentation:** No automated link between segmentation masks and clinical reports.
- **Safety Risks:** Standard LLMs lack evidence grounding, leading to hallucinations.

THE SOLUTION

An Adaptive, End-to-End Platform

- **Adaptive Segmentation Backbone:** Utilizing MoME+ to handle multi-modal inputs and robustly segment diverse tumor types.
- **Continual Learning Engine:** Implementing EWC & Replay Memory to expand model capabilities without performance degradation.

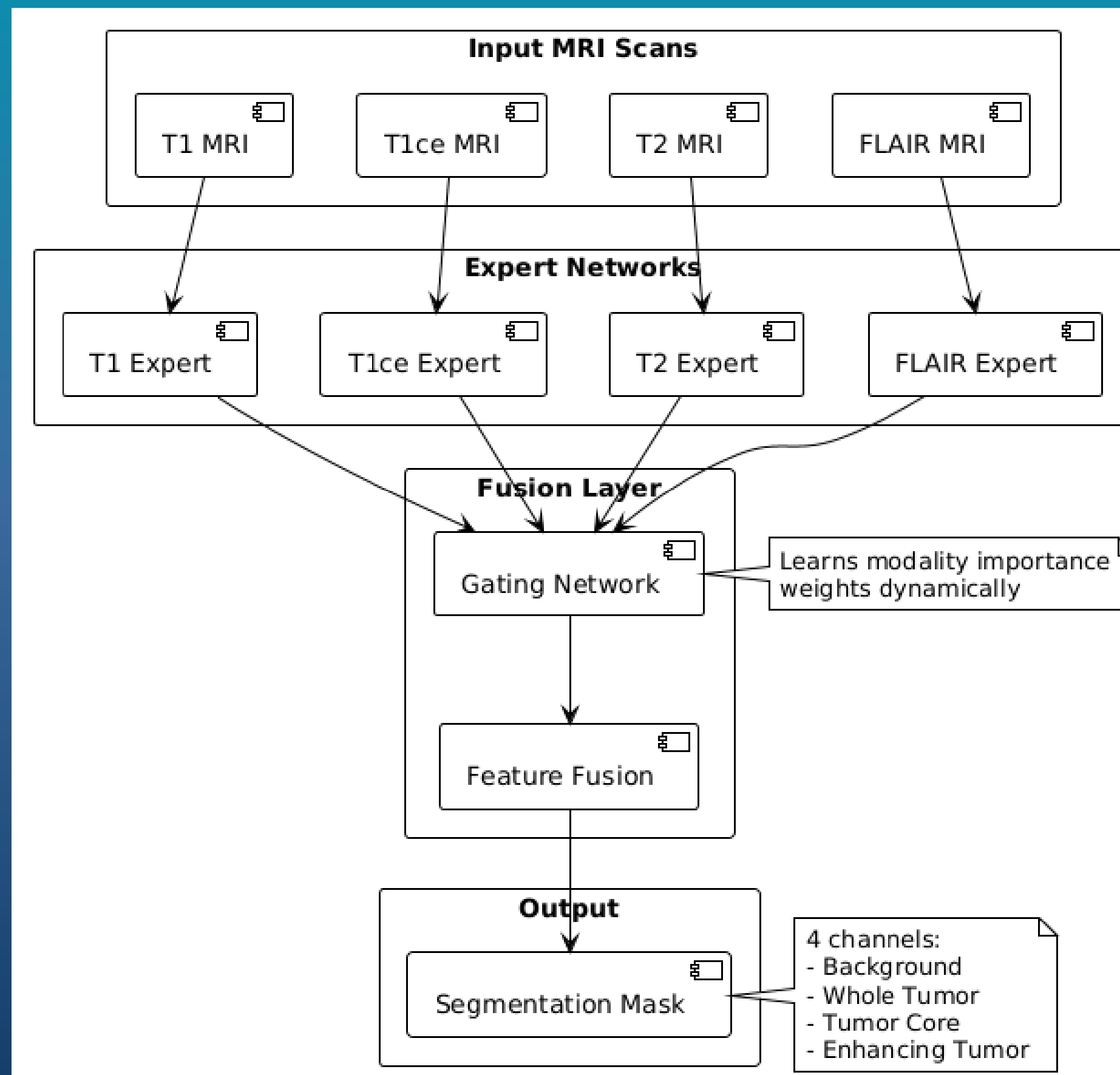


THE SOLUTION

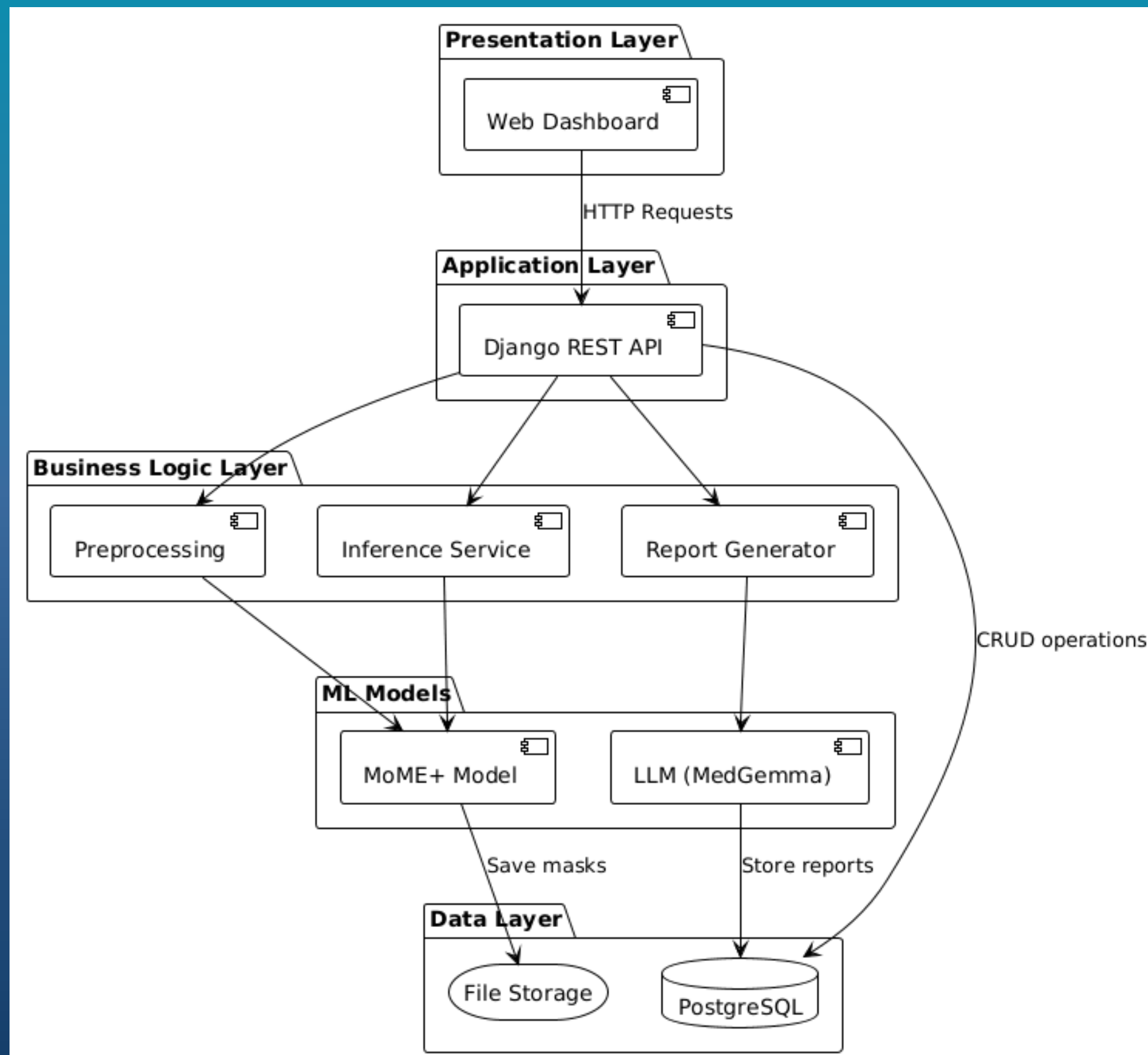
- **Unified Clinical Dashboard:**
A single interface bridging file upload, visual analysis, and report generation.
- **Evidence-Grounded Reporting:**
Generating reports strictly from extracted JSON metrics to ensure safety and eliminate hallucinations.



MOME+ ARCHITECTURE



SYSTEM ARCHITECTURE



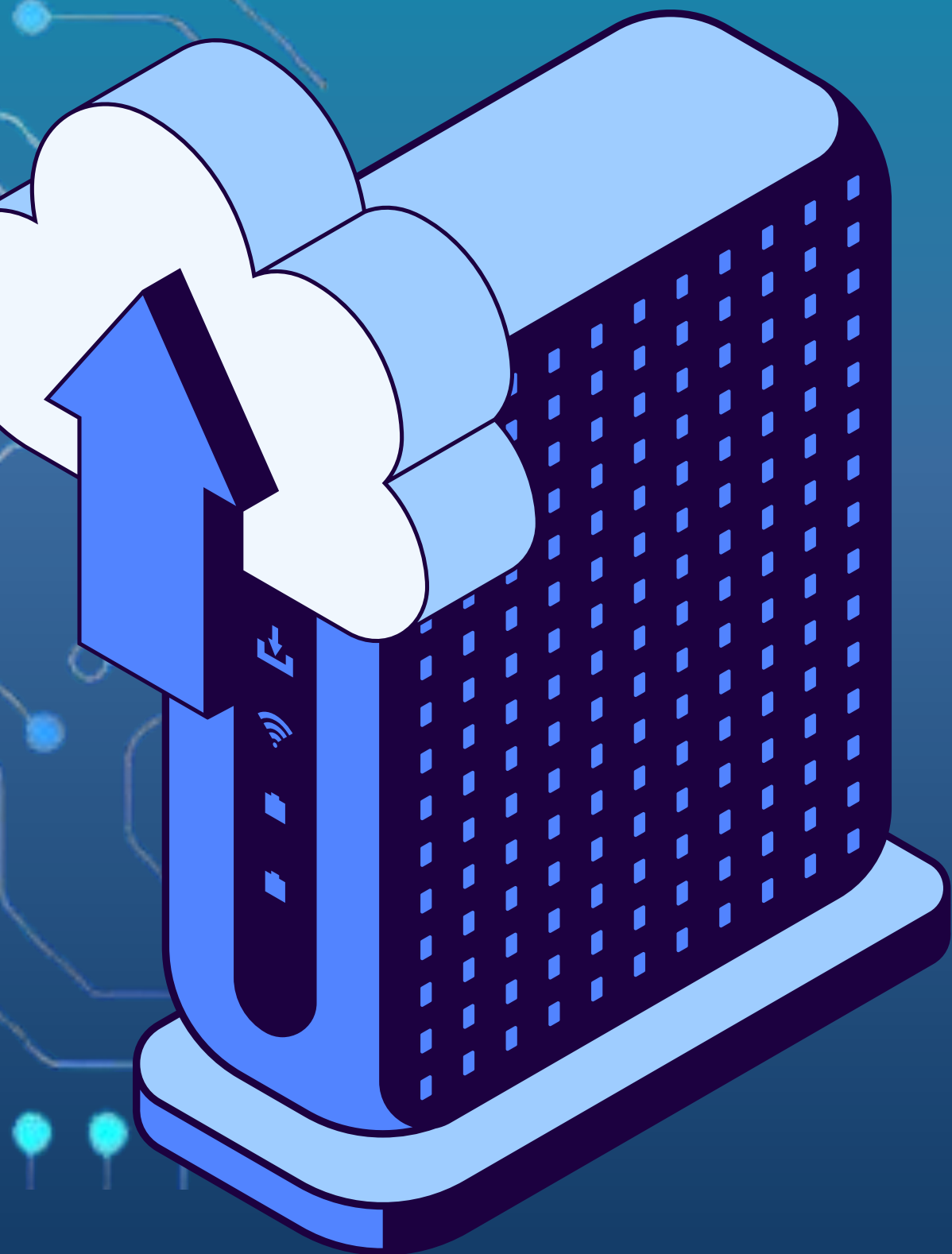


Photo Kiosk

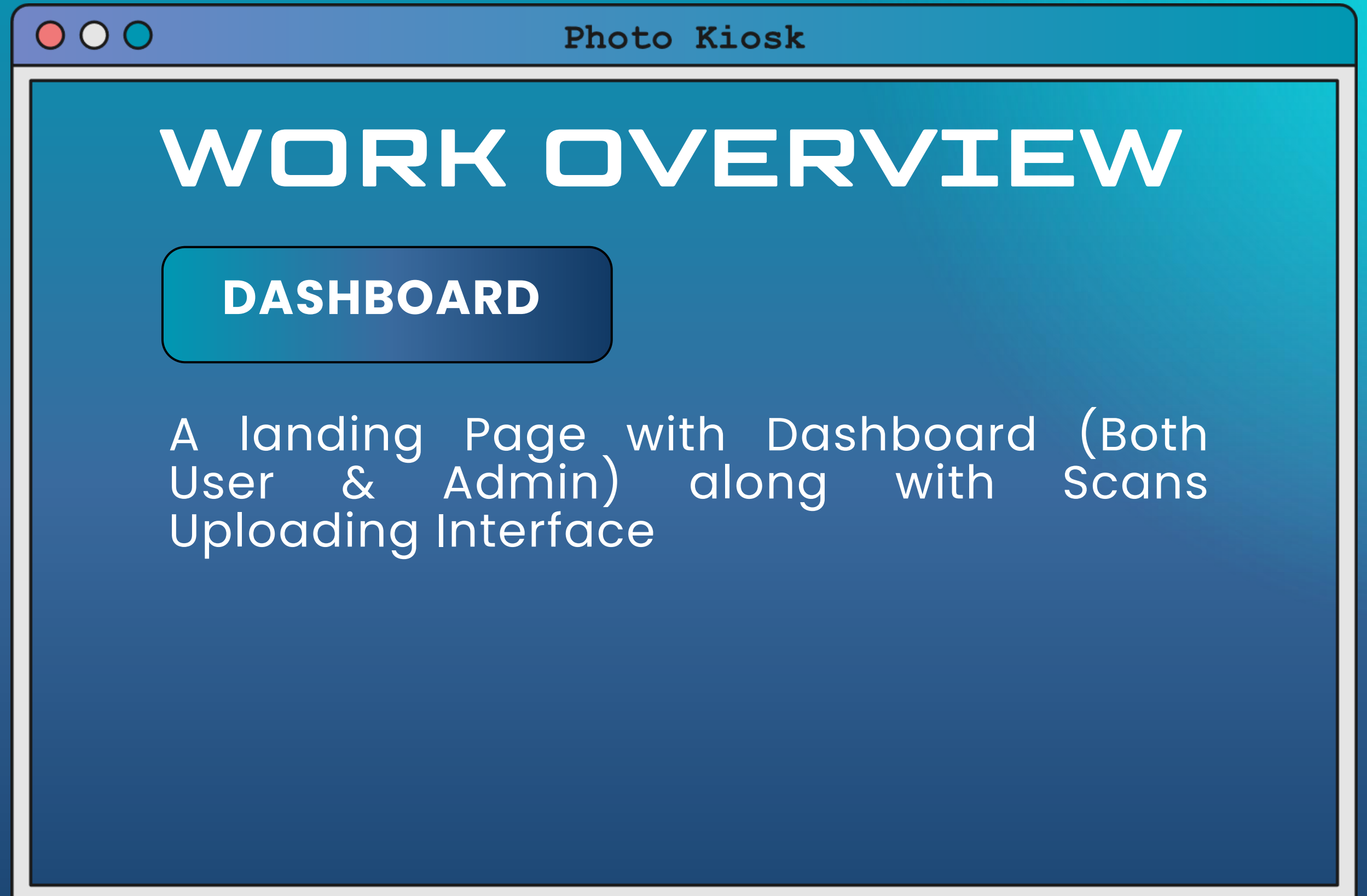
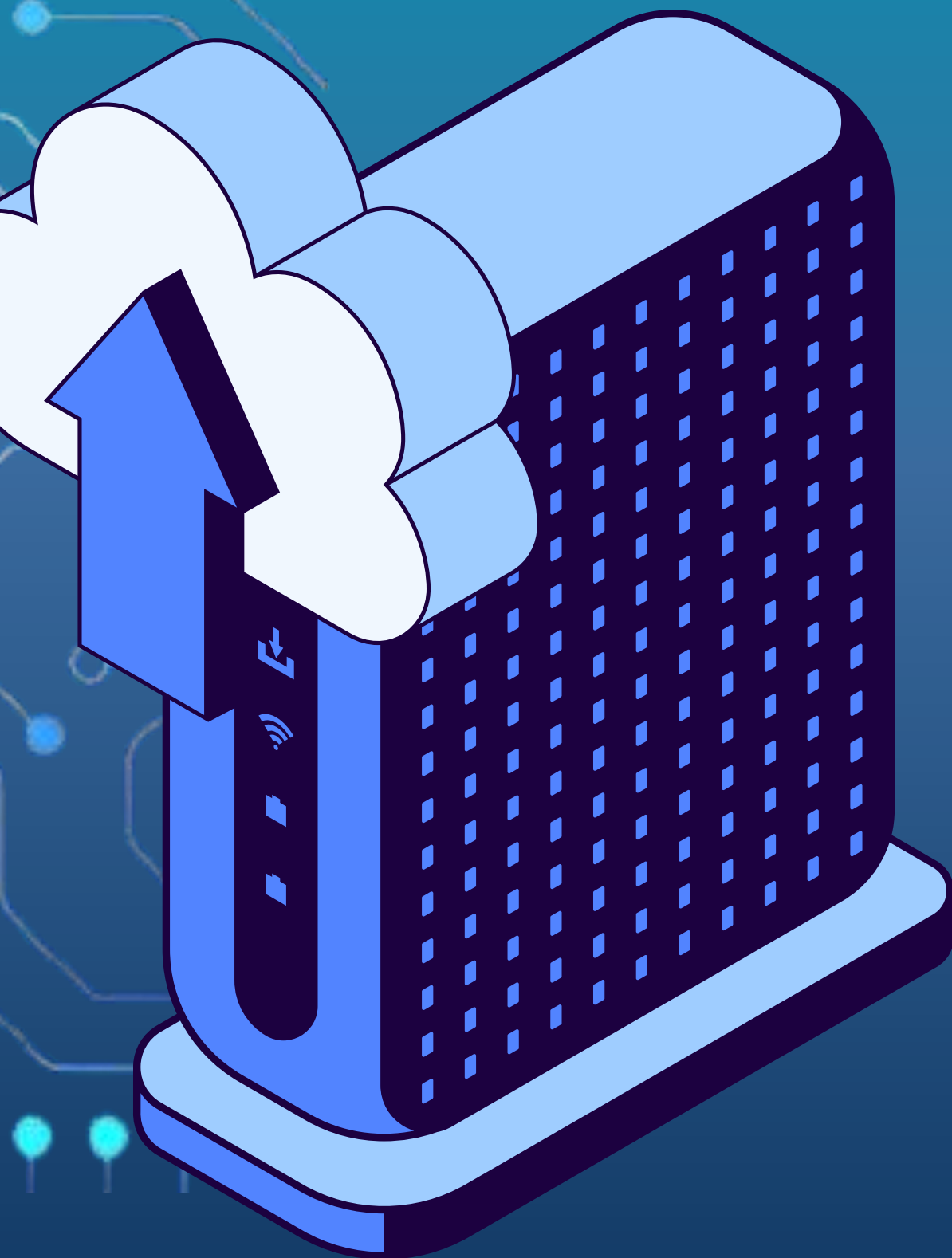
WORK OVERVIEW

SEGMENTATION ENGINE TRAINING

Four expert Unets (one for each modality) trained alongside the gating network using the Brats2024 BratsGLI dataset.

CONTINUAL TRAINING INTEGRATED

Fourth output class added (GTV masks) and model trained on Brats2024 BratsMEN dataset, retaining performance shown on BratsGLI while learning new features.



INITIAL TRAINING PROBLEMS

HARDWARE LIMITATIONS

- Initially training was done with preprocessing on-the-fly, leading to memory wastage.
 - Led to scrapping and restarting training from scratch, wasting about 100 hours of training.
- Training only allowed a batch size of 1 - 2, which lead to a degradation of performance.

NO VALIDATION MASKS

- The validation section of Brats datasets do not include ground truth masks
- This effectively reduced the amount of data we had for training.

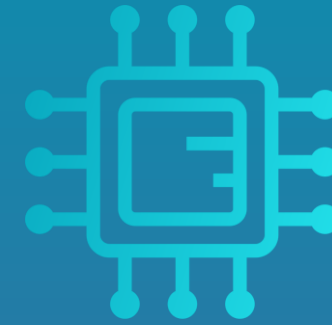
INITIAL TRAINING PROBLEMS

SOLUTIONS

- To address hardware limitations:
 - the use of FP16 variables was used, which was employed by using Automatic Mixed Precision (AMP).
 - Preprocessing data with 10 crops of 643 of each case.
- To introduce generality of model, only 6000 out of the 16000 total crops were considered each epoch, selected randomly.
- Validation data was split from training data with a 80-20 split.



MOME+ TRAINED

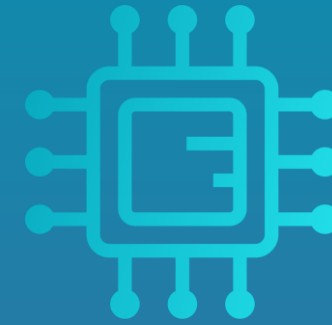


- After training adjustments, model was training for 80 epochs on BratsGLI dataset, producing a respectable **DICE score** of **0.8167**.





CONTINUAL LEARNING INTEGRATED



- Model was then further separately trained on BratsMEN dataset, which had a different tumor type, to preserve learnt features, Elastic Weight Consolidation and Replay Memory were employed.
- Results retained DICE score for GLI dataset at 0.79, while having a 0.60+ DICE on BratsMEN with 50 epochs of training.



LANDING PAGE



Sign In

Get Started

Research Prototype

AI-Powered Brain Tumor Segmentation & Analysis

Advanced medical AI platform for brain MRI analysis using mixture of modality experts. Automated segmentation, LLM-generated reports, and comprehensive visualization tools for healthcare professionals.

Get Started →

Sign In

 500+ Users •  10,000+ Cases Analyzed •  95% Accuracy

DASHBOARD

 Dashboard

 Cases

 Reports

 Settings

Dashboard

Overview of your medical AI analysis



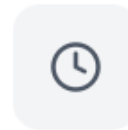
Total Cases
24



Completed
18



Processing
3



Pending
3

Recent Cases

[View All →](#)

CASE ID	AGE	SEX	STATUS	DATE	
BT-2024-001	58	M	completed	2024-12-18	View
BT-2024-002	42	F	processing	2024-12-18	View
BT-2024-003	65	M	completed	2024-12-17	View
BT-2024-004	51	F	pending	2024-12-17	View

 Logout

SCANS UPLOAD INTERFACE

 **MedicalAI**
Brain Segmentation

Dashboard

Cases

Reports

Settings

Logout

DR

Dr. Radiologist
Clinician

Upload MRI Scans

0/4 uploaded



Drop files or click to upload

Upload multiple files at once - we'll auto-detect modalities from filenames

OR UPLOAD INDIVIDUALLY



T1-weighted

Click to upload



T1-weighted Contrast Enhanced

Click to upload



T2-weighted

Click to upload



FLAIR

Click to upload

 Supported formats: NIfTI (.nii, .nii.gz). Maximum size: 500MB per file.
Tip: Include modality name in filename (e.g., patient_t1.nii.gz, case_t1ce.nii.gz)

MANAGEMENT TOOL - TRELLO

Trello

Q

Search

Create

14 days left

FK

FYP

000

FK

A

Share

Dashboard

→←

...

✓ System Architecture Design

✓ Basic Frontend

✓ Basic Backend

Admin Dashboard

Report Page

Inference Page

Update Model Page

+ Add a card

ML Model Training

→←

...

✓ Data Acquisition

✓ Data Preprocessing

✓ Baseline Segmentation Model Training

Continual Learning

+ Add a card

JSON Mapping and LLM Fine-Tuning

→←

...

Brain Atlas Mapping

JSON Generation

LLM Finetuning

+ Add a card

Completed

→←

...

+ Add a card

+ Add another list

Inbox

Planner

Board

Switch boards

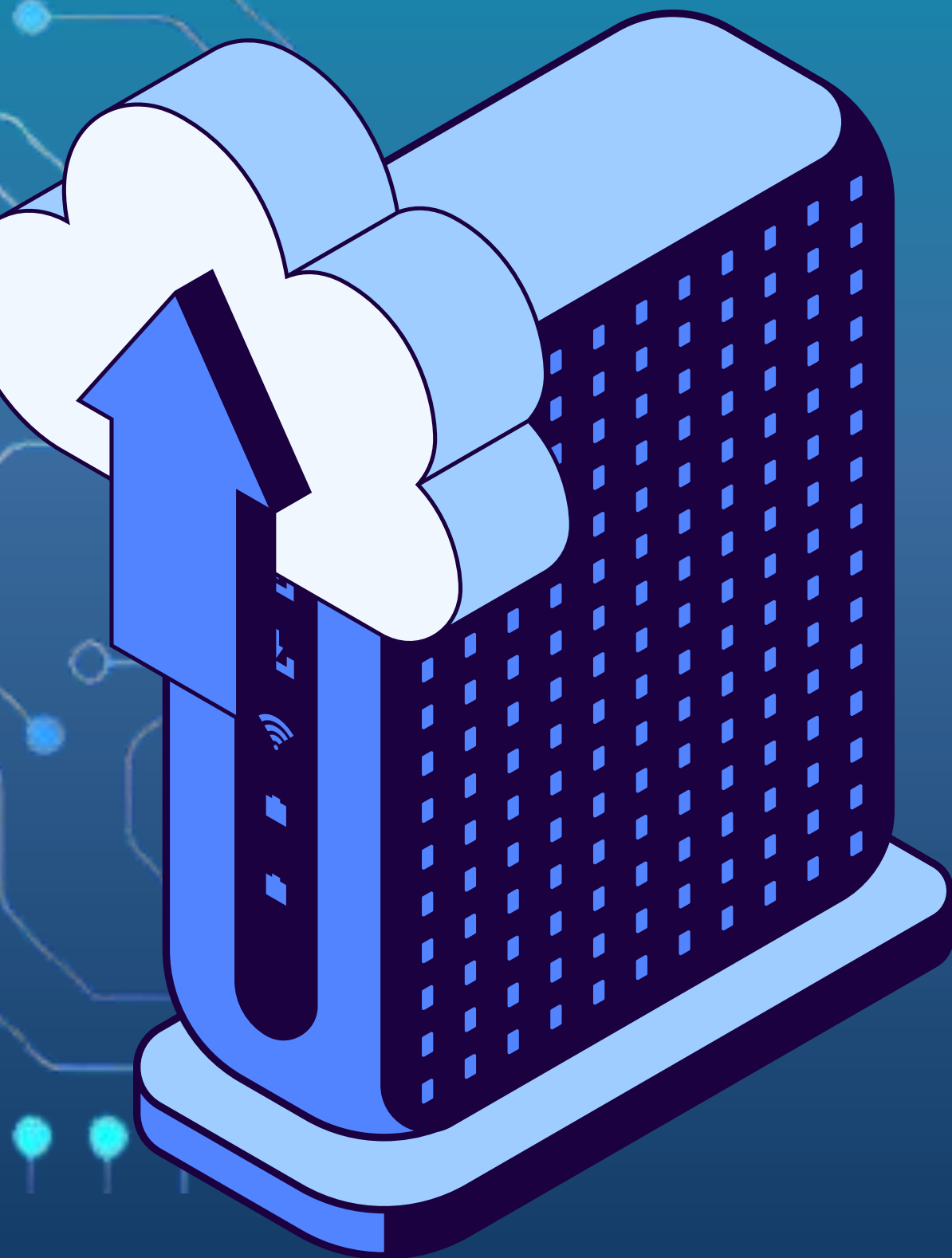


Photo Kiosk

FUTURE WORK

BRAIN ATLAS MAPPING

Mapping the outputs of the segmentation engine onto a 3D brain atlas.

LLM TRAINING

Generating ground-truth JSONs to train LLM for report generation.

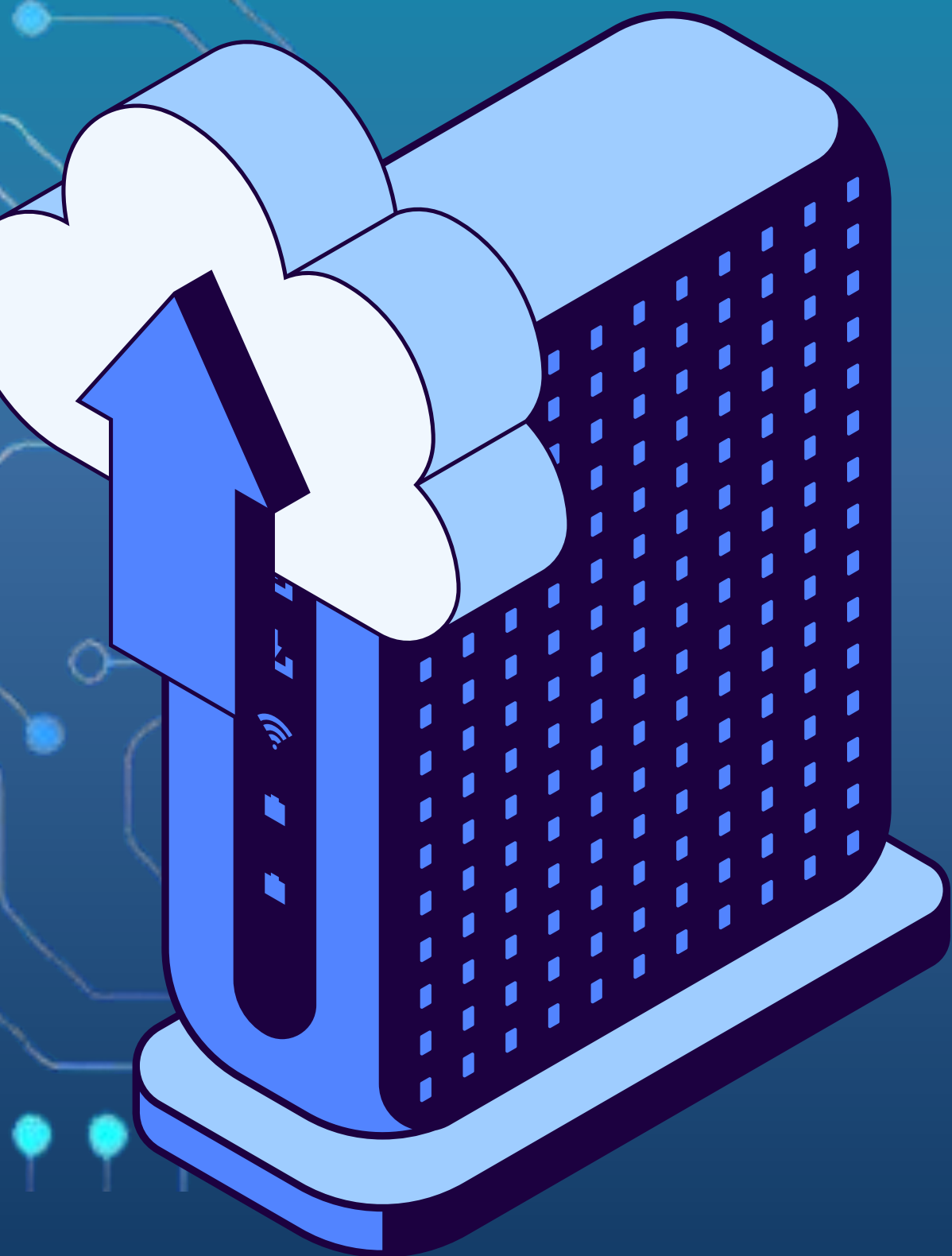


Photo Kiosk

FUTURE WORK

REFINE CONTINUAL LEARNING

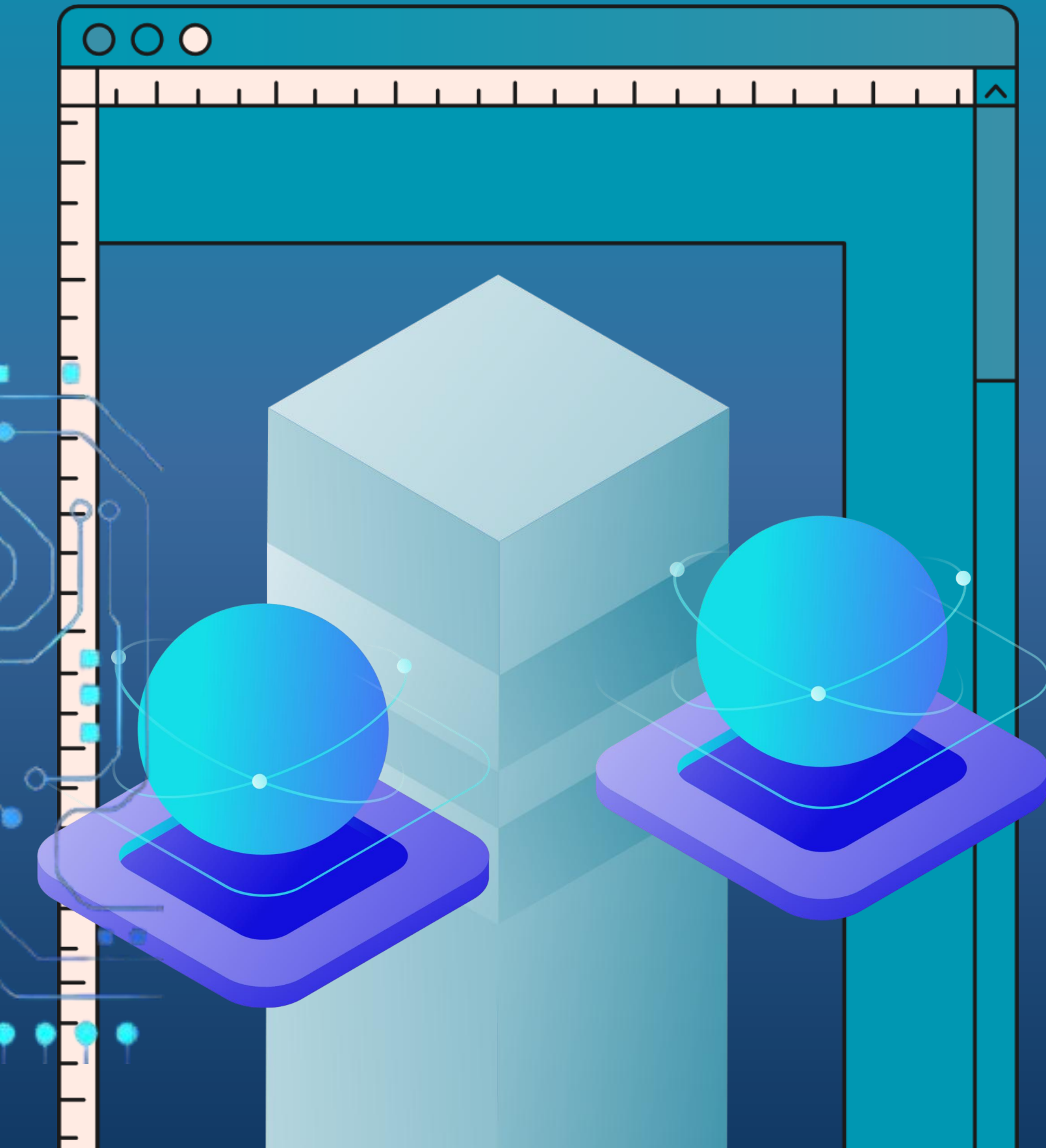
Quantify and minimize "forgetting" metrics during the GLI to MEN (and other datasets, such as Brain Metases) expansion by fine-tuning EWC and Replay Memory hyperparameters.

DASHBOARD VISUALIZATION

Implement 3D rendering for segmentation masks and finalize the interactive report editing interface.

CONCLUSION

- Successfully validated the adaptive segmentation pipeline on BraTS 2024 (GLI & MEN), achieving 0.8167 Dice while mitigating catastrophic forgetting.
- Delivered a functional End-to-End Dashboard prototype, proving the feasibility of integrating robust, continual-learning AI into a unified clinical workflow.





THANK YOU!